

Vpc challenge

VPC Transit Gateway & VPC Endpoint challenges

1. Design and deploy a scalable network architecture using AWS Transit Gateway to simplify network connectivity between multiple VPCs

Your VPCs (5) [Info](#)

Find VPCs by attribute or tag

Last updated 1 minute ago [Actions](#) [Create VPC](#)

☐	Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4 CIDR	IPv6 CIDR	DHCP option
☐	Vpc-2	vpc-02f873e8af148bd75	Available	–	–	Off	10.0.0.0/22	–	dopt-01024
☐	Vpc-03	vpc-0f4dba0516fbff538	Available	–	–	Off	10.0.0.0/21	–	dopt-01024
☐	–	vpc-00ce773080d81479f	Available	–	–	Off	172.31.0.0/16	–	dopt-01024
☐	Vpc-04	vpc-045d58cd7fa0dcc41	Available	–	–	Off	10.0.0.0/20	–	dopt-01024
☐	Vpc-1	vpc-0ecfc3f448cbf8029	Available	–	–	Off	10.0.0.0/24	–	dopt-01024

Select a VPC above

Subnets (4) [Info](#)

Find subnets by attribute or tag

Last updated less than a minute ago [Actions](#) [Create subnet](#)

☐	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR	IPv6 CIDR association ID
☐	Vpc-1-subnet	subnet-0396a64d89ce58fb0	Available	vpc-0ecfc3f448cbf8029 Vpc-1	Off	10.0.0.0/24	–	–
☐	Vpc-3-subnet	subnet-0f8592de90b131d4b	Available	vpc-0f4dba0516fbff538 Vpc-03	Off	10.0.0.0/24	–	–
☐	Vpc-2-subnet	subnet-0c89d8241c879636e	Available	vpc-02f873e8af148bd75 Vpc-2	Off	10.0.0.0/24	–	–
☐	Vpc-04-subnet	subnet-0bb7a3089e4580232	Available	vpc-045d58cd7fa0dcc41 Vpc-04	Off	10.0.0.0/24	–	–

Select a subnet

Transit gateway attachments (4) Info

Find transit gateway attachment by attribute or tag

1

Actions

Create transit gateway attachment

☐	Name	Transit gateway attachment ID	Transit gateway ID	State	Resource type	Resource ID	Asso
☐	Transit gateway atta...	tgw-attach-02603203d2260aec2	tgw-09231fedda4e90383	Available	VPC	ypc-0f4dba0516fbff538	tgw-
☐	Transit gateway atta...	tgw-attach-03bb33656553d756f	tgw-09231fedda4e90383	Available	VPC	ypc-02f873e8af148bd75	tgw-
☐	Transit gateway atta...	tgw-attach-0677411e21ac4fee	tgw-09231fedda4e90383	Available	VPC	ypc-0ecfc3f448cbf8029	tgw-
☐	Transit gateway atta...	tgw-attach-0d5a3b72be6759cd1	tgw-09231fedda4e90383	Available	VPC	ypc-045d58cd7fa0dccc41	tgw-

Transit gateway route tables (5) Info

Find transit gateway route table by attribute or tag

1

Actions

Create transit gateway route table

☐	Name	Transit gateway route table ID	Transit gateway ID	State	Default association route table	Default propagatio...
☐	Transit gateway rout...	tgw-rtb-0856bcc5b074042f9	tgw-09231fedda4e90383	Available	No	No
☐		tgw-rtb-0a2ca1bcbfce35d38	tgw-09231fedda4e90383	Available	Yes	Yes
☐	Transit gateway rout...	tgw-rtb-0c48fa21be6ec4265	tgw-09231fedda4e90383	Available	No	No
☐	Transit gateway rout...	tgw-rtb-0f52411759ad201fd	tgw-09231fedda4e90383	Available	No	No
☐	Transit Gateway rout...	tgw-rtb-0f5dedecc48fb92a4	tgw-09231fedda4e90383	Available	No	No

Select a transit gateway route table

Edit routes: rtb-0d466f78ac59a7141 / Routetable-1-Vpc-01

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/24	local	Active	No	CreateRouteTable
0.0.0.0/16	local	Active	No	CreateRoute
0.0.0.0/8	Transit Gateway	Active	No	CreateRoute
0.0.0.0/0	Transit Gateway	Active	No	CreateRoute

Add route

Cancel

Preview

Save changes

☐	-	rtb-0c8194809/b2e277e	-	-	Yes	vpc-045d58cd7fa0d41 Vpc-04	814787354428
☐	-	rtb-05feb239314df6325	-	-	Yes	vpc-0ecfc3f448cbf8029 Vpc-1	814787354428
☒	Routetable-1-Vpc-01	rtb-0d466f78ac59a7141	-	-	No	vpc-0ecfc3f448cbf8029 Vpc-1	814787354428
☐	Routetable-02-Vpc-02	rtb-0c126d1a79731fe2b	-	-	No	vpc-02f873e8af148bd75 Vpc-2	814787354428
☐	Routetable-03-Vpc-03	rtb-0580f1a390ed2efbc	-	-	No	vpc-0f4dba0516fbff538 Vpc-03	814787354428
☐	Routetable-04-Vpc-04	rtb-0340e9bf41828af73	-	-	No	vpc-045d58cd7fa0d41 Vpc-04	814787354428

rtb-0d466f78ac59a7141 / Routetable-1-Vpc-01

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/16	tgw-09231fedda4e90383	Active	No	Create Route
0.0.0.0/8	tgw-09231fedda4e90383	Active	No	Create Route
0.0.0.0/0	tgw-09231fedda4e90383	Active	No	Create Route
10.0.0.0/24	local	Active	No	Create Route Table

Instances (1/4) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set

Filter instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elas
☐ Ec2-Vpc-03	i-0414f80d117deb5cf	Running	t3.micro	Initializing	eu-north-1b	-	51.21.191.165	-
☒ Ec2-Vpc-01	i-07a1dbb87a0771e79	Running	t3.micro	3/3 checks passed	eu-north-1b	-	-	-
☐ Ec2-Vpc-02	i-01aeaa0bef3b0bc31	Running	t3.micro	3/3 checks passed	eu-north-1a	-	51.20.181.136	-
☐ Ec2-Vpc-04	i-02f7ac7ab3b69bffe	Running	t3.micro	Initializing	eu-north-1a	-	-	-

i-07a1dbb87a0771e79 (Ec2-Vpc-01)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Instance summary

Instance ID

[i-07a1dbb87a0771e79](#)

IPv6 address

-

Hostname type

IP name: ip-10-0-0-222.eu-north-1.compute.internal

Public IPv4 address

-

Instance state

Running

Private IP DNS name (IPv4 only)

[ip-10-0-0-222.eu-north-1.compute.internal](#)

Private IPv4 addresses

[10.0.0.222](#)

Public DNS

-

Feedback

Console Mobile App

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2. Configure VPC endpoints to securely access AWS services without Internet Gateways or NAT Gateways.

Instances (1/4) Info

Last updated 9 minutes ago

Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set

Filter instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elas
☐ vpc2-ec2	i-031b9219d3f028d55	Running	t3.micro	3/3 checks passed	eu-north-1b	-	13.60.74.137	-
☒ vpc1-ec1	i-0ff1907883991bbfd	Running	t3.micro	Initializing	eu-north-1b	-	-	-
☐ public-ec2	i-05dfbc8446088cf7	Running	t3.micro	3/3 checks passed	eu-north-1b	-	-	-
☐ vpc3-ec3	i-0229987b5ab8c996f	Running	t3.micro	Initializing	eu-north-1a	-	13.62.51.150	-

i-0ff1907883991bbfd (vpc1-ec1)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

☐ ⓘ ↗

Your VPCs

VPCsVPC encryption controls

Your VPCs (4) Info

Last updated 10 minutes ago

Actions

Create VPC

Q Filter VPCs by attribute or tag

☐	Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4 CIDR	II
☐	Vpc-02	vpc-0e3b1980a8d2e1845	Available	–	–	Off	10.0.1.0/28	–
☐	Vpc-03	vpc-077a6a05c79b82c5b	Available	–	–	Off	10.0.2.0/28	–
☐	Vpc-public	vpc-00ce773080d81479f	Available	–	–	Off	172.31.0.0/16	–
☐	Vpc-01	vpc-0f89ace8ac1917695	Available	–	–	Off	10.0.0.0/28	–

Select a VPC above

☐ ⓘ ↗

Updated routes for rtb-0ef0713b480631fa1 / vpc3-rt-3 successfully

Details

Route tables (1/4) Info

Last updated 1 minute ago

Actions

Create route table

Q Find route tables by attribute or tag

☐	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
☐	vpc1-rt-1	rtb-017be52c41746b477	–	–	Yes	vpc-0f89ace8ac1917695 Vpc-01	814787354428
☐	vpc2-rt-2	rtb-0746c57bcffb99236	–	–	Yes	vpc-00ce773080d81479f Vpc-...	814787354428
☒	vpc3-rt-3	rtb-0ef0713b480631fa1	–	–	Yes	vpc-077a6a05c79b82c5b Vpc-...	814787354428
☐	default-routetable	rtb-0073a13a088c9a1d1	–	–	Yes	vpc-0e3b1980a8d2e1845 Vpc-...	814787354428

rtb-0ef0713b480631fa1 / vpc3-rt-3

☐ ⓘ ↗

Subnets (4) Info

Last updated 9 minutes ago

Actions

Create subnet

Q Find subnets by attribute or tag

☐	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR
☐	Private-subnet-02	subnet-0757719c25b036fe9	Available	vpc-0e3b1980a8d2e1845 Vpc-...	Off	10.0.1.0/28	–
☐	Default-subnet	subnet-0b701bac630b1d817	Available	vpc-00ce773080d81479f Vpc-...	Off	172.31.0.0/18	–
☐	subnet-03	subnet-0710d0096baeadf0e	Available	vpc-077a6a05c79b82c5b Vpc-...	Off	10.0.2.0/28	–
☐	private-subnet-01	subnet-0c86184f65da2d8ee	Available	vpc-0f89ace8ac1917695 Vpc-01	Off	10.0.0.0/28	–

Select a subnet

Transit gateway attachments (4) Info

Find transit gateway attachment by attribute or tag

< 1 >

Actions

Create transit gateway attachment

☐	Name	Transit gateway attachment ID	Transit gateway ID	State	Resource type	Resource ID	Association route
☐	vpc3-transit gateway...	tgw-attach-081897bed8d15f60b	tgw-09231fedda4e90383	Available	VPC	vpc-077a6a05c79b82c5b	tgw-rtb-0a2ca1bc
☐	Test-transit-attachm...	tgw-attach-0b7655263151c8473	tgw-09231fedda4e90383	Available	VPC	vpc-00ce773080d81479f	tgw-rtb-0a2ca1bc
☐	vpc1-transit gateway...	tgw-attach-0bcc56024275792b8	tgw-09231fedda4e90383	Available	VPC	vpc-0f89ace8ac1917695	tgw-rtb-0a2ca1bc
☐	vpc2-transit gateway...	tgw-attach-0d9a3d6c181b0b92d	tgw-09231fedda4e90383	Available	VPC	vpc-0e3b1980a8d2e1845	tgw-rtb-0a2ca1bc

Select a transit gateway attachment

Transit gateways (1) Info

Find transit gateway by attribute or tag

< 1 >

Actions

Create transit gateway

☐	Name	Transit gateway ID	Owner ID	State
☐	Transit-gateway-01	tgw-09231fedda4e90383	814787354428	Available

Transit gateway route tables (5) Info

Find transit gateway route table by attribute or tag

< 1 >

Actions

Create transit gateway route table

☐	Name	Transit gateway route table ID	Transit gateway ID	State	Default association route table	Default propagati...
☐	Transit gateway rout...	tgw-rtb-0856bcc5b074042f9	tgw-09231fedda4e90383	Available	No	No
☐		tgw-rtb-0a2ca1bcbf3e35d38	tgw-09231fedda4e90383	Available	Yes	Yes
☐	Transit gateway rout...	tgw-rtb-0c48fa21be6ec4265	tgw-09231fedda4e90383	Available	No	No
☐	Transit gateway rout...	tgw-rtb-0f52411759ad201fd	tgw-09231fedda4e90383	Available	No	No
☐	Transit Gateway r...	tgw-rtb-0f5dedecd48fb92a4	tgw-09231fedda4e90383	Available	No	No

Select a transit gateway route table

```
[ec2-user@ip-172-31-21-119 ~]$ ping www.google.com
PING www.google.com (142.251.153.119) 56(84) bytes of data.
64 bytes from 142.251.153.119 (142.251.153.119): icmp_seq=1 ttl=118 time=3.21 ms
64 bytes from 142.251.153.119 (142.251.153.119): icmp_seq=2 ttl=118 time=3.23 ms
64 bytes from 142.251.153.119 (142.251.153.119): icmp_seq=3 ttl=118 time=3.22 ms
64 bytes from 142.251.153.119 (142.251.153.119): icmp_seq=4 ttl=118 time=3.23 ms
64 bytes from 142.251.153.119 (142.251.153.119): icmp_seq=5 ttl=118 time=3.22 ms
64 bytes from 142.251.153.119 (142.251.153.119): icmp_seq=6 ttl=118 time=3.23 ms
64 bytes from 142.251.153.119 (142.251.153.119): icmp_seq=7 ttl=118 time=3.22 ms
64 bytes from 142.251.153.119 (142.251.153.119): icmp_seq=8 ttl=118 time=3.23 ms
64 bytes from 142.251.153.119 (142.251.153.119): icmp_seq=9 ttl=118 time=3.24 ms
64 bytes from 142.251.153.119 (142.251.153.119): icmp_seq=10 ttl=118 time=3.26 ms
64 bytes from 142.251.153.119 (142.251.153.119): icmp_seq=11 ttl=118 time=3.23 ms
64 bytes from 142.251.153.119 (142.251.153.119): icmp_seq=12 ttl=118 time=3.24 ms
64 bytes from 142.251.153.119 (142.251.153.119): icmp_seq=13 ttl=118 time=3.25 ms
64 bytes from 142.251.153.119 (142.251.153.119): icmp_seq=14 ttl=118 time=3.23 ms
64 bytes from 142.251.153.119 (142.251.153.119): icmp_seq=15 ttl=118 time=3.24 ms
64 bytes from 142.251.153.119 (142.251.153.119): icmp_seq=16 ttl=118 time=3.22 ms
64 bytes from 142.251.153.119 (142.251.153.119): icmp_seq=17 ttl=118 time=3.23 ms
64 bytes from 142.251.153.119 (142.251.153.119): icmp_seq=18 ttl=118 time=3.32 ms
64 bytes from 142.251.153.119 (142.251.153.119): icmp_seq=19 ttl=118 time=3.22 ms
64 bytes from 142.251.153.119 (142.251.153.119): icmp_seq=20 ttl=118 time=3.23 ms
64 bytes from 142.251.153.119 (142.251.153.119): icmp_seq=21 ttl=118 time=3.23 ms
64 bytes from 142.251.153.119 (142.251.153.119): icmp_seq=22 ttl=118 time=3.32 ms
64 bytes from 142.251.153.119 (142.251.153.119): icmp_seq=23 ttl=118 time=3.24 ms
64 bytes from 142.251.153.119 (142.251.153.119): icmp_seq=24 ttl=118 time=3.23 ms
64 bytes from 142.251.153.119 (142.251.153.119): icmp_seq=25 ttl=118 time=3.23 ms
```

```
_/m/'
Last login: Mon Oct 5 17:04:56 2026 from 13.48.4.203
[ec2-user@ip-172-31-21-119 ~]$ ping 172.31.43.196
PING 172.31.43.196 (172.31.43.196) 56(84) bytes of data.
64 bytes from 172.31.43.196: icmp_seq=176 ttl=127 time=0.253 ms
64 bytes from 172.31.43.196: icmp_seq=177 ttl=127 time=0.200 ms
64 bytes from 172.31.43.196: icmp_seq=178 ttl=127 time=0.190 ms
64 bytes from 172.31.43.196: icmp_seq=179 ttl=127 time=0.215 ms
64 bytes from 172.31.43.196: icmp_seq=180 ttl=127 time=0.224 ms
64 bytes from 172.31.43.196: icmp_seq=181 ttl=127 time=0.213 ms
64 bytes from 172.31.43.196: icmp_seq=182 ttl=127 time=0.211 ms
64 bytes from 172.31.43.196: icmp_seq=183 ttl=127 time=0.217 ms
64 bytes from 172.31.43.196: icmp_seq=184 ttl=127 time=0.193 ms
64 bytes from 172.31.43.196: icmp_seq=185 ttl=127 time=0.207 ms
64 bytes from 172.31.43.196: icmp_seq=186 ttl=127 time=0.197 ms
64 bytes from 172.31.43.196: icmp_seq=187 ttl=127 time=0.200 ms
64 bytes from 172.31.43.196: icmp_seq=188 ttl=127 time=0.221 ms
64 bytes from 172.31.43.196: icmp_seq=189 ttl=127 time=0.204 ms
64 bytes from 172.31.43.196: icmp_seq=190 ttl=127 time=0.208 ms
64 bytes from 172.31.43.196: icmp_seq=191 ttl=127 time=0.198 ms
64 bytes from 172.31.43.196: icmp_seq=192 ttl=127 time=0.205 ms
64 bytes from 172.31.43.196: icmp_seq=193 ttl=127 time=0.204 ms
64 bytes from 172.31.43.196: icmp_seq=194 ttl=127 time=0.181 ms
64 bytes from 172.31.43.196: icmp_seq=195 ttl=127 time=0.199 ms
64 bytes from 172.31.43.196: icmp_seq=196 ttl=127 time=0.200 ms
64 bytes from 172.31.43.196: icmp_seq=197 ttl=127 time=0.201 ms
64 bytes from 172.31.43.196: icmp_seq=198 ttl=127 time=0.198 ms
64 bytes from 172.31.43.196: icmp_seq=199 ttl=127 time=0.165 ms
64 bytes from 172.31.43.196: icmp_seq=200 ttl=127 time=0.199 ms
64 bytes from 172.31.43.196: icmp_seq=201 ttl=127 time=0.198 ms
64 bytes from 172.31.43.196: icmp_seq=202 ttl=127 time=0.217 ms
64 bytes from 172.31.43.196: icmp_seq=203 ttl=127 time=0.203 ms
```

